Laser Cutting, Bending and Fabrication Design Guidelines:-

1. Aluminium Material Selection for Bending

- Aluminium grades such as **Al 6061, Al 6082, Al 6063, Commercial Aluminium, and Al 7075** are **not suitable for bending above 3 mm thickness** due to their higher hardness and lower ductility.

- For these grades, it is **recommended to limit bending operations to thicknesses below 3 mm** to avoid cracking or deformation during bending.

- If bending is required for thicker sections, alternate materials should be considered.

2. Aluminium 1110 Material for Bending

- **Aluminium 1110** has better ductility compared to other aluminium grades.

- This material **can be bent effectively up to 10 mm thickness** without significant cracking.

- For parts requiring **higher thickness bending**, Aluminium 1110 is the **preferred material**.

- Welding operations are also **more effective and reliable in Aluminium 1110** compared to other aluminium grades.

3. Welding Considerations for Aluminium

- Aluminium grades such as **6061, 6082, 6063, and 7075** are **not ideal for welding**, particularly for fabrication involving frequent bending and forming.

- **Aluminium 1110** is comparatively **better suited for welding operations** due to its higher weldability.

4. Stainless Steel (SS) and Mild Steel (MS) Bending Considerations

- Stainless Steel and Mild Steel with **higher thicknesses may face bending challenges**, particularly when:
    - The **component size is small**
    - The **bend height is insufficient**
- To improve bending quality in thick SS and MS parts:
    - Maintain **higher bend height**
    - Increase the **bend radius**

5. Bend Radius for Thick Materials

- For materials with **higher thickness**, the **bend radius must be increased** to prevent cracking and deformation.
- Proper bend radius improves:
    - Structural integrity
    - Dimensional accuracy
    - Surface finish

6. Hole and Slot Placement Near Bend Areas

- For thicker materials, **holes and slots must be positioned at a sufficient distance from the bending edge**.
- If holes or slots are located **too close to the bend line**:
    - The holes may **distort or deform**
    - The bending operation may **damage the part**

- Proper clearance from the bend edge ensures **clean and accurate bending**.

7. Minimum Hole Size for Laser Cutting

- The **hole diameter should be at least 2 times the material thickness** for proper laser cutting.

Example:

- 2 mm thickness → Minimum hole size **4 mm**
- 3 mm thickness → Minimum hole size **6 mm**

If holes are smaller than this guideline:

- The **laser cutting process may damage the hole geometry**
- The **part quality may be compromised**

8. Tapping Hole Guidelines

- If the part requires **tapping (threading)**:
    - **Avoid cutting the final tapping hole directly using laser cutting.**
- Laser cutting may cause:
    - Slight **hole taper**
    - **Surface hardness**
    - **Irregular hole geometry**
- This may result in **bolt slipping during assembly**.

Recommended Practice

- The laser cut a **smaller pilot hole**.

- Perform **drilling to the final tapping size** before threading.
- This ensures:
    - Accurate thread formation
    - Proper bolt engagement
    - Reliable assembly performance.

Manufacturing Guidelines for Milling Operations

1. Material Selection Considerations

- Different materials require **specific cutting parameters and tooling selection** for effective milling operations.
- Materials such as **Aluminium, Mild Steel (MS), Stainless Steel (SS), Brass, and Copper** have different machinability characteristics.
- Aluminium alloys generally allow **higher cutting speeds and better machinability**.
- Stainless Steel requires **lower cutting speeds and rigid tooling** due to work hardening properties.
- Hard materials require **coated carbide tools or special inserts** to prevent excessive tool wear.

---

2. Tool Selection

- Proper tool selection is critical for **surface finish, dimensional accuracy, and tool life**.
- Common milling tools include:
    - **End Mills**
    - **Face Mills**
    - **Ball Nose Cutters**
    - **Slot Drills**
- Use **carbide tools for harder materials** and **HSS tools for softer materials or low production volumes**.

- Tool diameter should be selected based on **slot width, pocket size, and corner radius requirements**.

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3. Minimum Corner Radius

- Internal corners produced by milling **cannot be perfectly sharp** due to the round geometry of cutting tools.
- The **minimum internal radius should be equal to or greater than the cutter radius**.

Example:

- If using **Ø10 mm end mill**, minimum internal radius = **5 mm**
- If sharp internal corners are required:
    - Secondary machining processes such as **EDM or wire cutting** may be necessary.

---

4. Depth of Cut Considerations

- Excessive depth of cut may result in:
    - Tool breakage
    - Poor surface finish
    - Increased vibration
- Recommended depth of cut should depend on:
    - Tool diameter
    - Material hardness

    - Machine rigidity

General guideline:

- Depth of cut should **not exceed 50–70% of tool diameter** in most milling operations.

---

5. Slot Milling Guidelines

- Full-width slot milling creates **higher tool load and heat generation**.
- For deeper slots:
    - Use **multiple passes instead of single deep cuts**.
    - Maintain proper chip evacuation.
- Slots deeper than **3× tool diameter** may require special tooling strategies.

---

6. Wall Thickness in Machined Parts

- Thin walls may deform during machining due to cutting forces.
- Recommended minimum wall thickness:

Material	Minimum Wall Thickness
Aluminium	1 mm
Steel	1.5 mm

Stainless Steel 2 mm

- Thin features require **reduced cutting forces and optimized tool paths**.

---

7. Hole Machining Guidelines

- Holes produced by drilling or milling should follow proper **diameter-to-depth ratios**.
- Recommended maximum hole depth:
- Standard drilling: **Up to 5× hole diameter**
- Deep hole drilling: **Up to 10× hole diameter with special tools**
- For precision holes:
    - Use **remaining operations after drilling**.

---

8. Tapping Guidelines

- Holes intended for tapping should follow **standard tapping drill sizes**.
- Recommended practice:
    - Drill using **standard tapping drill diameter**
    - Perform **thread tapping operation separately**
- Avoid:
    - Oversized holes

    - Laser-cut holes for tapping
    - Irregular hole geometry

This ensures:

- Proper thread engagement
- Strong bolt fastening
- Reliable assembly.

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9. Surface Finish Considerations

Surface finish depends on:

- Cutting speed
- Feed rate
- Tool sharpness
- Tool path strategy

Recommended practices:

- Use a finishing **pass after rough machining**.
- Maintain **proper coolant supply**.
- Use **high-speed finishing tools** for better surface quality.

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10. Tool Path Optimization

- Efficient tool paths improve:
    - Machining time
    - Tool life
    - Surface finish

Recommended strategies:

- **Climb milling** for better finish and lower tool wear.
- **Conventional milling** for roughing in some materials.
- Avoid sudden tool direction changes.

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11. Workholding and Fixture Design

- Proper clamping is critical to avoid:
    - Part vibration
    - Dimensional inaccuracies
    - Tool breakage

Recommended practices:

- Use **rigid fixtures and clamps**.
- Ensure **proper part support for thin components**.
- Avoid excessive clamping forces that may deform the part.

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12. Coolant and Chip Removal

- Coolant improves:
    - Tool life
    - Surface finish
    - Heat dissipation

Chip removal is important to prevent:

- Tool breakage
- Surface damage
- Poor machining quality

Use:

- **Flood coolant**
- **Air blast**
- **Chip evacuation systems**

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13. Dimensional Tolerance Considerations

- Very tight tolerances increase:
    - Machining time
    - Production cost
    - Inspection requirements

General tolerance guideline:

Feature	Recommended Tolerance
General machining	±0.1 mm
Precision machining	±0.02 mm
High precision	±0.005 mm

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14. Common Milling Defects and Causes

Defect	Possible Cause
Poor surface finish	Dull tool, high feed rate
Tool breakage	Excessive depth of cut
Chatter marks	Machine vibration
Dimensional errors	Improper fixture
Burr formation	High feed or worn tool

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15. Inspection and Quality Control

- All machined components should be verified using:
    - **Vernier calipers**
    - **Micrometers**
    - **Height gauges**
    - **CMM inspection for precision parts**
- Ensure:
    - Dimensional accuracy
    - Surface finish quality
    - Proper thread formation

Drilling and Tapping Guidelines for Industrial Manufacturing

1. General Design Considerations for Drilling

1. Ensure **proper material selection** before drilling (Aluminium, Mild Steel, Stainless Steel, Brass, etc.).

2. Maintain **minimum edge distance** to avoid cracking or deformation.

    - Recommended: **Edge distance ≥ 1.5 × hole diameter**.

3. Maintain **minimum hole-to-hole spacing**:

    - Recommended: **≥ 2 × hole diameter**.

4. Use **center drilling or spot drilling** before deep drilling for accurate hole positioning.

5. For holes deeper than **3 × diameter**, use **peck drilling** to prevent tool breakage and chip accumulation.

6. Ensure proper **coolant or cutting fluid** is used to reduce heat and improve tool life.

7. Maintain **perpendicularity of drill** to the work surface for accurate hole geometry.

8. For thin sheet metal, use **step drills or backing plates** to avoid burrs and distortion.

9. Use the remaining **operation** if a tight tolerance hole is required.

10. Always provide **chamfer (0.5–1 mm × 45°)** at hole entry to assist tapping.

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2. Tapping Design Considerations

1. Ensure **correct tap drill size** before tapping.

2. Recommended **thread engagement**:

    - Aluminium: **1–1.5 × thread diameter**
    - Steel: **1 × thread diameter**
    - Cast iron: **1–1.2 × thread diameter**
3. Maintain **minimum wall thickness**:
    - ≥ **1.5 × thread diameter**
4. For **blind holes**, provide extra depth:
    - **Thread depth + 1.5 × pitch**
5. Use **proper lubrication** during tapping to avoid tap breakage.
6. Avoid tapping in **laser-cut holes without re-drilling** because taper and HAZ may affect thread quality.
7. For **thin sheet metal**, use:
    - **Nutserts / rivet nuts**
    - **Thread forming inserts**
8. Use **spiral flute taps** for blind holes and **spiral point taps** for through holes.
9. Ensure **tap alignment** with drill axis.
10. For high production, use **machine tapping instead of manual tapping** for consistency.

---

3. Standard Tap Drill Size Chart (Metric Threads)

Tap Size	Pitch (mm)	Drill Size (mm)
M2	0.4	1.6
M2.5	0.45	2.05
M3	0.5	2.5
M4	0.7	3.3
M5	0.8	4.2
M6	1.0	5.0
M8	1.25	6.8
M10	1.5	8.5
M12	1.75	10.2
M14	2.0	12.0
M16	2.0	14.0
M18	2.5	15.5

M20	2.5	17.5
M22	2.5	19.5
M24	3.0	21.0

---

4. Recommended Hole Depth for Tapping

Tap Size	Minimum Hole Depth (mm)
M3	6–8
M4	8–10
M5	10–12
M6	12–15
M8	16–20
M10	20–25
M12	25–30

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5. Drilling Speed Guidelines (General)

Material	Recommended Speed
Aluminium	High speed
Mild Steel	Medium speed
Stainless Steel	Low speed
Brass	Medium-high speed

---

6. Common Problems and Solutions

Problem	Cause	Solution
Tap breakage	Small drill size	Use correct tap drill
Poor thread finish	No lubrication	Use tapping oil
Oversize thread	Drill size too large	Maintain standard drill

Burr formation	High feed rate	Reduce feed and deburr

---

7. Industry Best Practices

1. Always **deburr drilled holes** before tapping.
2. Use **quality HSS or carbide taps** for long tool life.
3. Maintain **tool inspection schedule**.
4. Maintain **coolant filtration system** in CNC machines.
5. For aluminium parts, **thread forming taps** give stronger threads.
6. Always verify **thread using GO/NO-GO gauges**.

Industrial Drilling and Tapping Guidelines

(Metric, BSP, and NPT Threads – Manufacturing Reference)

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1. General Design Considerations for Tapping

1. Always select the **correct tap drill size** based on thread standard and pitch.
2. Maintain proper **thread engagement length** for structural strength.
3. Avoid tapping directly in **laser-cut holes** without re-drilling.
4. Always provide **lead-in chamfer (0.5–1 mm × 45°)** before tapping.
5. Maintain **minimum edge distance**:

Minimum edge distance ≥ **1.5 × thread diameter**

6. Maintain **hole spacing**:

Minimum hole spacing ≥ **2 × thread diameter**

7. For **blind holes**, provide additional clearance depth:

Hole depth = Thread depth + **1 to 1.5 × pitch**

8. Always use **coolant or tapping oil** during tapping.
9. Ensure **tap alignment** with the hole axis.
10. For thin sheet metals (<3 mm), avoid direct tapping; instead use:

- Rivet nuts
- Weld nuts
- Thread inserts

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2. Metric Thread Tap Drill Size Chart

Tap Size	Pitch	Drill Size	Recommended Thread Depth
M2	0.4	1.6 mm	3–4 mm
M2.5	0.45	2.05 mm	4–5 mm
M3	0.5	2.5 mm	5–6 mm
M4	0.7	3.3 mm	6–8 mm
M5	0.8	4.2 mm	8–10 mm
M6	1.0	5.0 mm	10–12 mm
M8	1.25	6.8 mm	12–16 mm
M10	1.5	8.5 mm	15–20 mm

M12	1.75	10.2 mm	18–25 mm
M14	2.0	12 mm	20–28 mm
M16	2.0	14 mm	24–32 mm
M18	2.5	15.5 mm	28–36 mm
M20	2.5	17.5 mm	30–40 mm
M22	2.5	19.5 mm	32–45 mm
M24	3.0	21 mm	36–50 mm

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3. BSP (British Standard Pipe) Tap Drill Sizes

BSP threads are **parallel threads commonly used in hydraulic and pneumatic systems**.

BSP Size	Threads per Inch	Tap Drill Size	Recommended Thread Depth
1/16 BSP	28	7.7 mm	8–10 mm
1/8 BSP	28	8.8 mm	10–12 mm

1/4 BSP	19	11.8 mm	12–15 mm
3/8 BSP	19	15.2 mm	14–18 mm
1/2 BSP	14	18.6 mm	16–22 mm
3/4 BSP	14	24.1 mm	18–25 mm
1 BSP	11	30.5 mm	22–30 mm
1¼ BSP	11	39 mm	28–35 mm
1½ BSP	11	44.5 mm	30–40 mm
2 BSP	11	56 mm	35–45 mm

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4. NPT (National Pipe Thread) Tap Drill Sizes

NPT threads are **tapered threads used for sealing in piping systems**.

NPT Size	Threads per Inch	Tap Drill Size	Recommended Thread Depth
1/16 NPT	27	6 mm	6–8 mm

1/8 NPT	27	8.6 mm	8–10 mm
1/4 NPT	18	11.2 mm	10–14 mm
3/8 NPT	18	14.7 mm	12–16 mm
1/2 NPT	14	18.6 mm	14–20 mm
3/4 NPT	14	24.5 mm	16–22 mm
1 NPT	11.5	30 mm	18–25 mm
1¼ NPT	11.5	38.5 mm	20–30 mm
1½ NPT	11.5	44 mm	22–35 mm
2 NPT	11.5	56 mm	25–40 mm

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5. Recommended Thread Engagement (Industry Standard)

Material	Thread Engagement

Aluminium	1.5 × diameter
Mild Steel	1 × diameter
Stainless Steel	1–1.25 × diameter
Cast Iron	1–1.5 × diameter

Example:

M10 tap in steel

Minimum thread depth = **10 mm**

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6. Types of Taps Used in Industry

1. Hand Tap Set

Used for manual tapping.

Types include:

1. **Taper Tap** – initial threading
2. **Intermediate Tap** – medium cutting
3. **Bottoming Tap** – used in blind holes

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2. Spiral Flute Tap

Used for **blind holes**.

Advantages:

- Removes chips upward
- Prevents chip clogging

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3. Spiral Point Tap (Gun Tap)

Used for **through holes**.

Advantages:

- Pushes chips forward
- Ideal for CNC tapping

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4. Form Tap (Roll Tap)

Used mainly in **aluminium and ductile materials**.

Advantages:

- Stronger threads
- No chips produced

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5. Pipe Taps

Used for **BSP and NPT threads** in fluid systems.

Types:

- NPT Tap

- BSP Tap
- BSPT Tap

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7. Drilling Speed Recommendations

Material	Speed	Lubrication
Aluminium	High	Light oil
Mild Steel	Medium	Cutting oil
Stainless Steel	Low	Heavy cutting oil
Brass	Medium-high	Dry or light oil

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8. Tapping Problems and Solutions

Problem	Cause	Solution
Tap breakage	Incorrect drill size	Use proper tap drill
Rough threads	No lubrication	Apply tapping oil

Loose threads	Drill size too large	Reduce drill diameter
Tap stuck in hole	Chip accumulation	Use peck tapping

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9. Quality Inspection Methods

1. Use **GO / NO-GO thread gauges**
2. Inspect threads visually for burrs
3. Measure **thread depth using depth gauge**
4. Perform **leak test for pipe threads**

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10. Best Practices for Manufacturing

1. Always **spot drill before drilling**.
2. Deburr holes before tapping.
3. Maintain proper **coolant flow**.
4. Use **CNC tapping cycles for production parts**.
5. Use **thread inserts for high-load applications**.

CNC Lathe (Turning) Manufacturing Guidelines

1. Material Machinability Guidelines

Different materials require different cutting parameters, tooling, and machining strategies.

Aluminium (6061, 6082, 6063)

- Very good machinability.
- Allows **high cutting speeds and feed rates**.
- Produces good surface finish.
- Requires **sharp carbide tools with positive rake**.
- Chip evacuation must be controlled to avoid built-up edge.

Mild Steel (MS)

- Moderate machinability.
- Requires **coated carbide inserts**.
- Use coolant to reduce heat generation.
- Chips may become continuous and require chip breaking tools.

Stainless Steel (SS304, SS316)

- Difficult to machine due to **work hardening**.
- Requires **lower cutting speed and higher rigidity**.
- Use **coated carbide inserts with positive rake geometry**.
- Continuous coolant flow is recommended.

Brass

- Excellent machinability.
- Allows **very high cutting speeds**.
- Usually machined without coolant.
- Produces short chips and good surface finish.

Copper

- Soft and ductile material.
- Chips tend to stick to cutting tool.
- Use **sharp tools and lower feed rates**.

Engineering Plastics (Nylon, Delrin, PTFE)

- Easy to machine.
- Requires **very sharp tools**.
- Avoid excessive heat generation.

---

2. Diameter and Length Feasibility

Length to Diameter Ratio

Long parts tend to deflect during machining.

Recommended ratio:

- **Up to 3:1 → No support required**
- **3:1 to 6:1 → Tailstock support recommended**

- **Above 6:1 → Steady rest required**

Example:

- Diameter = 20 mm
- Maximum unsupported length ≈ 60 mm

---

3. Tool Selection

Common Turning Tools

- Rough turning inserts
- Finishing inserts
- Boring bars
- Grooving tools
- Parting tools
- Threading inserts

Insert Material

Tool Type	Application
Carbide	High production machining
Coated Carbide	Steel and stainless steel

HSS	Soft materials or low production

---

4. Depth of Cut Guidelines

Depth of cut depends on:

- Material hardness
- Machine rigidity
- Tool geometry

Typical guidelines:

Roughing:

- 1 mm to 5 mm depth of cut

Finishing:

- 0.2 mm to 0.5 mm

Excessive depth may cause:

- Tool breakage
- Vibration
- Poor surface finish

---

5. Surface Finish Guidelines

Surface finish depends on:

- Feed rate
- Tool nose radius
- Cutting speed
- Machine stability

Typical Ra values:

Operation	Surface Finish
Rough turning	3.2 – 6.3 µm
Semi finishing	1.6 – 3.2 µm
Finishing	0.8 – 1.6 µm

Lower feed rate improves surface finish.

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6. Minimum Wall Thickness

Thin parts deform easily during machining.

Recommended minimum thickness:

Material	Minimum Wall Thickness

Aluminium	1 mm
Mild Steel	1.5 mm
Stainless Steel	2 mm

For thin walls:

- Reduce cutting force
- Use finishing passes only

---

7. Internal Boring Guidelines

Internal machining requires enough space for boring bars.

Recommended bore diameter:

Minimum bore diameter ≥ **1.5 × boring bar diameter**

Small bore sizes cause:

- Tool vibration
- Poor surface finish
- Tool breakage

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8. Grooving and Parting

Grooving

- Groove width must match tool width.
- Deep grooves require **multiple passes**.

Parting

- Tool must be aligned with **centerline of spindle**.
- Use coolant to reduce heat.
- Slow feed rate prevents tool breakage.

---

9. Thread Manufacturing

Threads can be produced using:

- Single point threading
- Tapping
- Die threading
- Thread milling (in CNC)

Thread relief should be provided.

Typical relief size:

- Width = **1 to 1.5 × thread pitch**

---

10. Drilling on Lathe

Steps for accurate drilling:

1. Center drilling
2. Pilot drilling
3. Final drilling
4. Reaming (if precision required)

Depth guidelines:

Maximum drilling depth ≈ **5 × drill diameter**

Deep holes require:

- Peck drilling
- Coolant supply

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11. Chamfer and Radius Recommendations

Sharp edges should be avoided.

Standard practice:

- Chamfer = **0.5 mm to 1 mm**
- Fillet radius = **0.5 mm minimum**

Benefits:

- Safe handling
- Reduced burr formation
- Easier assembly

---

12. Tolerance Guidelines

Feature	Recommended Tolerance
General machining	±0.1 mm
Precision turning	±0.02 mm
High precision	±0.005 mm

Tighter tolerance increases:

- Machining time
- Tool wear
- Production cost

---

13. Workholding Methods

3-Jaw Chuck

- Quick setup
- Suitable for round parts

4-Jaw Chuck

- Used for irregular shapes
- Higher accuracy

Collet Chuck

- Used for small diameter parts
- High precision

Mandrels

- Used for internal holding of parts

---

14. Coolant and Chip Control

Coolant improves:

- Tool life
- Surface finish
- Heat removal

Types of cooling:

- Flood coolant
- Air blast
- Minimum quantity lubrication (MQL)

Proper chip control prevents:

- Surface scratches
- Tool damage

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15. Common Lathe Machining Defects

Defect	Cause
Chatter marks	Machine vibration
Poor surface finish	Worn insert
Dimensional error	Improper clamping
Burr formation	Aggressive cutting parameters

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16. Inspection and Quality Control

Inspection tools:

- Vernier caliper
- Micrometer
- Bore gauge
- Thread gauge
- Dial indicator
- CMM inspection

Verify:

- Diameter accuracy

- Concentricity
- Surface finish
- Thread quality

---

17. Design for Manufacturability (DFM)

To reduce machining cost:

- Avoid very tight tolerances unless necessary
- Use standard thread sizes
- Avoid deep narrow grooves
- Maintain proper chamfers
- Provide sufficient tool access